



Assessing the Impact of Various Drivers on Ground Water Resources in WB

An interactive data visualization dashboard built on raw and synthetic GWL data

INTRODUCTION

Groundwater is one of the most important sources of fresh water. It supports farming, drinking water supply, and many industries. In India, a large share of irrigation and daily water use depends on groundwater. But in recent years, rapid urban growth, heavy water extraction, land-use change, and climate variation have caused groundwater levels to fall in many areas.

PROBLEM STATEMENT

West Bengal has been experiencing increasing groundwater depletion across several districts, driven by a combination of climate variability, land-use and land-cover changes, and growing water demand. Changes in rainfall and temperature can affect natural recharge, while rapid urbanization and agricultural expansion alter infiltration and runoff patterns. At the same time, excessive groundwater extraction for agriculture and domestic use is intensifying stress on available resources.

OBJECTIVES

- Develop an integrated 2 pages Dashboard to assess the impacts of climate variability and land-use changes on groundwater resources.
- Analyze long-term groundwater level trends across 23 selected districts of West Bengal.
- Investigate pre-monsoon and post-monsoon groundwater level variations and seasonal trends.
- Evaluate the relationship between rainfall, groundwater levels and other LULC Drivers.
- Helps users to identify Land use, Mean depth, Rainfall detection across multiple districts and seasons.

DATASET DESCRIPTION

“This dashboard uses a mixed dataset combining raw and synthetic records. The dataset includes Date, Year, District, Season, MeanDepth, Rainfall, TMax, TMin, Cropland, BuiltUp, WaterBody, Forest, and Other LULC. It is designed for exploratory analysis, pattern detection, and visualization of groundwater and land-use trends across districts and seasons.”

DASHBOARD FEATURES

- **KPI Cards** — KPI cards for total districts, Years covered, and mean depth average level.
- **Interactive Map** — District-wise Mean Depth distribution shown geographically.
- **Dual Bar Chart** — Mean Depth vs Rainfall Comparison.
- **Clustered Bar Chart** — Ground Water depth changes across seasons and districts.
- **Circle Chart** — Represented Avg Rainfall and Mean Depth Relationship using Trend Line.
- **Stacked Bar Chart** — Represented Land use using Various drivers such as Builtup, Waterbody, Forest, and Other LULC.
- **Ranking Tables** — Top 5 Districts by Deep Mean Depth, and Top 5 Districts by Shallow Mean Depth.
- **Filters** — Drill-down District, Season, and Date.

EXPECTED OUTCOMES

- Simplifies interpretation of complex raw synthetic data for a general audience.
- Enables meaningful exploratory analysis and patterns detection of Groundwater across West Bengal Districts.
- Supports academic research, classroom learning, and policy planning.
- Demonstrates practical application of BI and data visualization techniques.

CONCLUSION

This project shows that groundwater conditions in West Bengal vary significantly across districts due to a combination of rainfall, temperature, seasonal changes, and land-use patterns. Districts with higher groundwater depth and weaker rainfall response appear more vulnerable and may need priority attention.

Link: https://public.tableau.com/views/WB_GWL_Analysis/WB_GWL_OVERVIEW?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link